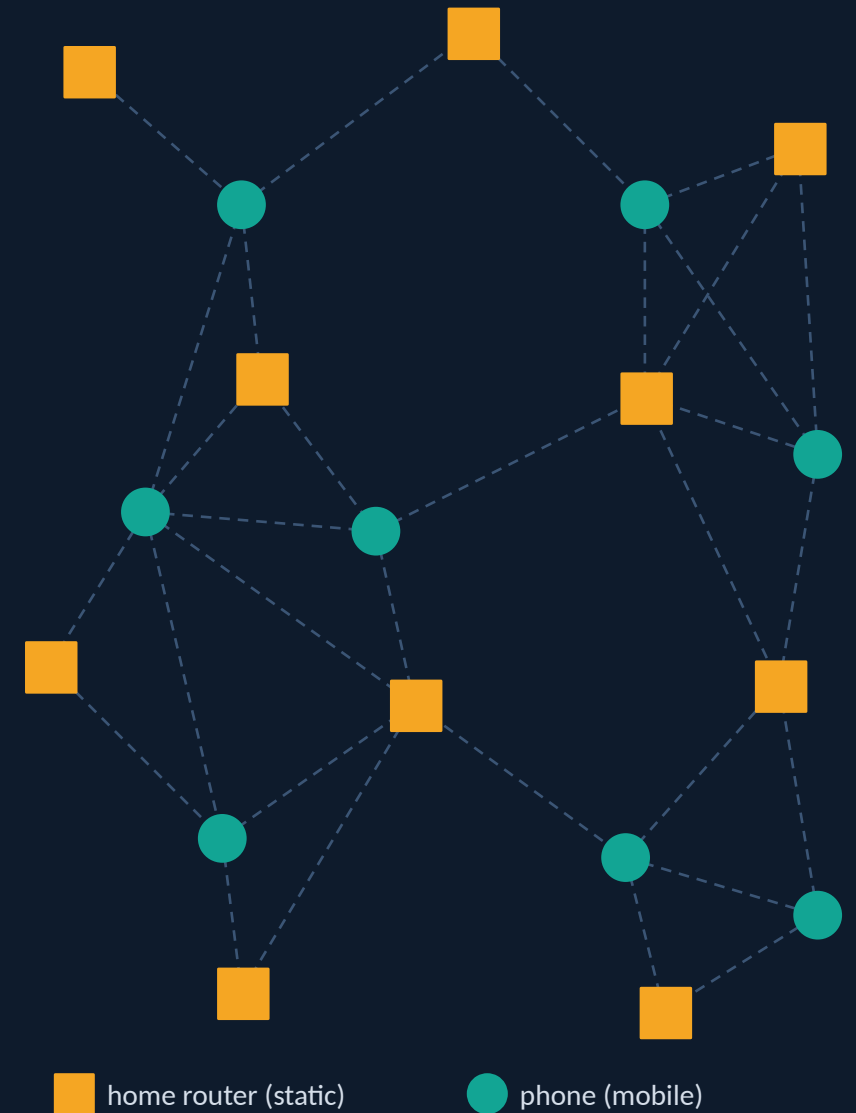


Pomóc

The network that is already there

A dual-use civic mesh: neighbourhood resource sharing on an ordinary day, resilient emergency communication when mobile networks and power fail. Same identity, same protocol, same phones and home routers.



Cities hold the resources and lose the route to them



An ordinary Tuesday

resources idle because nobody nearby knows

- Tools, parking spots and skills sit unused next to people who need them
- Every small need becomes a trip to a shop
- Neighbourhood platforms are centralised and have no reason to work offline



The day the network goes dark

help is nearby but unreachable

- Official warnings (Alert RCB) ride on cellular networks and fail with the cells
- Consumer mesh apps are installed after the crisis starts and carry no identity
- A stranger's request cannot be verified, so nobody can safely act on it

Both are routing problems on the same graph: who is physically near whom.

One network, one identity, two rule sets

	Peace mode (default)	Emergency mode
Purpose	Waste reduction, local utilisation	Resilient communication, life-critical routing
Messages	Lend, borrow, give, sell, information	Life-critical, safety, official alerts, "I am OK" check-ins
Payments	Optional micro-payments or credits	Disabled
Reach	Hop-limited: default 3, max 6	6 to 15 hops by level, store-and-forward
Transport	Peers over mesh, internet only to authorities	Router mesh, then phone-only mesh
Trigger	Default	Backhaul loss (L1) or signed declaration



Routers: backbone

Static, powered, dense in cities, already remotely managed by ISPs. Relay only.



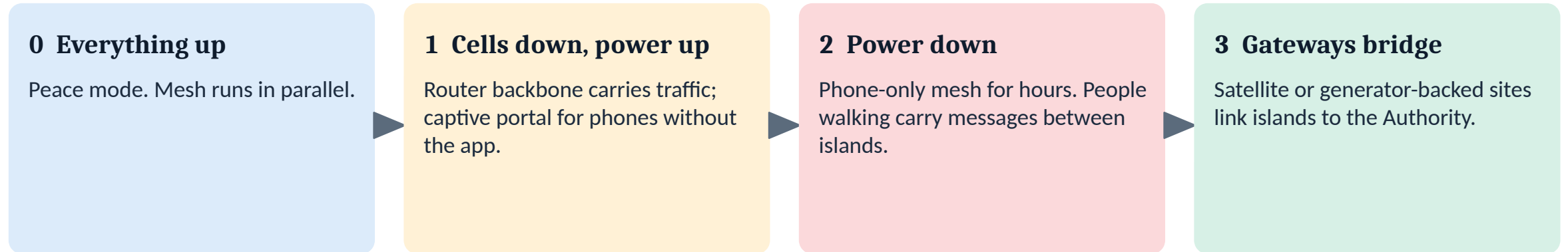
Phones: workers

Mobile and battery-powered: the only thing left when the power goes. Ask, offer, relay.

Why the state: only a state can verify identity, enable ISP relays and sign declarations everyone accepts.

Inspired by an ant colony: no node sees the whole, each acts on local information, coordination emerges.

When the network dies, the crowd keeps routing



Scenario (illustrative): cardiac arrest, the nearest public AED is 4 hops away

- 1**
Ola's phone sends a signed LIFE_CRITICAL request at top priority
- 2**
Mesh floods it with de-duplication, pre-empting every other class
- 3**
The citizen nearest the registered AED accepts: "fetching AED"
- 4**
Per-hop instructions move the AED hand to hand along the found path
- 5**
Assumed outcome: AED arrives in about 90 s, against about 6 min for staff on site

The network routes the message, the crowd routes the object.

Emergency policy is a signed, regional preset

	L1 Disruption	L2 Disaster	L3 Security
Typical cause	Cell outage, cable cut, cyberattack	Flood, storm, long blackout	Terror, war, hybrid attack
Entered by	Local automation or declaration	Declaration only	Declaration only
Router portal	Read + check-in, aggregated	+ structured request, flagged unverified	Read only, one-way
Citizen classes	Life-critical, safety, check-in, info, free give	Same as L1	Life-critical, safety, check-in. Info off
Hop limit	10	15	6
Phone topology gossip	On	On	Off
Radio emission	Normal	Normal	Reduced duty cycle

Declarations are regional (one district at L3, the rest at L2). Downgrades need a signed all-clear or expiry; L3 always steps down through L1.

Built to resist abuse

Jamming gains nothing. Local automation reaches L1 at most; L2 and L3 need an Authority signature.

Mesh load is bounded. A router emits at most one signed PORTAL_SUMMARY per minute, so load scales with routers, not attackers.

L3 is stricter. Citizen INFO off stops rumours; phone gossip off hides where people are.

Peacetime is why it is installed before the crisis



Everyday value

- Borrow a drill, get sour cream from a neighbour, hand over a parking spot
- Hop limit 3 keeps it hyperlocal by protocol
- Optional credits: a tool pays itself back through community use



A reason to keep it

- Verified neighbours make lending to strangers safe
- Bounded notifications: quiet hours, mutes, hop caps
- AED, EpiPen and insulin requests work in peace too
- Peerby reports about 80% of requests filled within 30 minutes



Value for the city

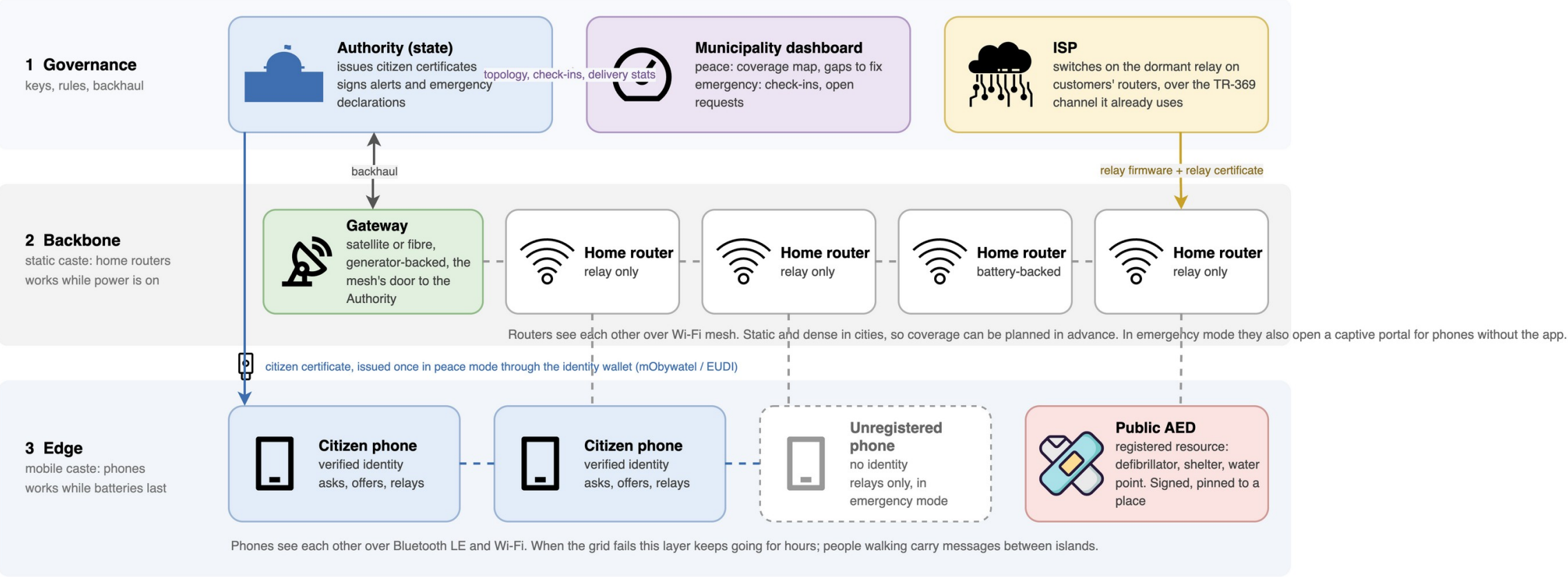
- Predictable graph: coverage gaps, articulation points, percolation threshold
- Shows where battery-backed relays or gateways belong
- Open data per district: transactions, waste avoided

Same identity, same graph, same protocol: the peacetime habit becomes the emergency capability.

Architecture: citizens ask, routers and gateways carry

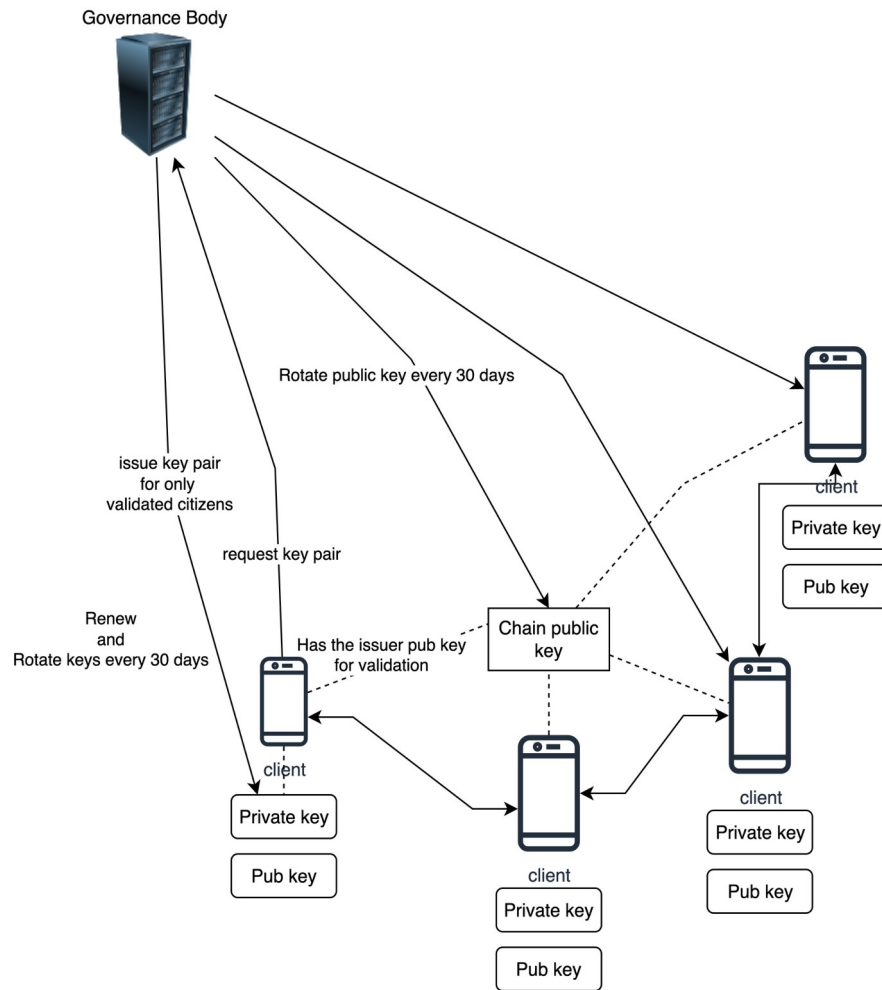
Pomóc

The network that is already there. One protocol, one identity, two rule sets: peace and emergency.



Reading the picture: dashed line = radio neighbours, an edge of the proximity graph · solid arrow = administrative or backhaul relationship · blue = verified citizen · dashed box = no identity, relay only. Only citizens can ask; routers and gateways only carry.

Trust model: verify anything, offline



Illustrative key flow; keys live in the phone secure element.

Issuer-generated keys. The Authority HSM issues key pair and certificate after wallet-verified registration, wrapped to the phone's secure element.

Offline verification. Any node checks any message against the Authority root key embedded in the app. No network round trip.

“Routers cannot act” is cryptographic. Relay certificates grant forwarding only; a request signed by one is dropped everywhere.

Forged alerts die. Declarations and alerts use separate Authority keys; unverifiable ones are dropped by every node.

Short-lived. Target 30-day keys renewed silently, over the mesh when offline; target 90-day emergency grace so an outage never locks anyone out.

Honest trade-off. The Authority could impersonate citizens. Bounded by no key retention, k-of-n escrow, dual-control HSMs and a public transparency log.

Technical feasibility: every block has a precedent

Block	What we rely on	Closest precedent
Router relay	ISP remote management (TR-069 / TR-369) on OpenWrt, prpIOS, RDK-B	00000JAPAN since 2014; Comcast opened 141k to 261k hotspots before hurricanes
Phone mesh	BLE and Wi-Fi Aware / Direct; messages sized for one BLE advertisement (target 240 B)	Relay-by-Smartphone (Tohoku), Bridgefy 12M+ users, bitchat
Offline trust	ed25519 signatures, certificate chain against embedded root, secure-element keys	Standard PKI, HSMs, Android Keystore / iOS Secure Enclave
Identity	National identity wallet: mObywatel, EUDI wallet	Alert RCB, announced move into mObywatel (2025)
Alerts	CAP payload so existing origination tools feed the mesh	Regulatory basis: EECC Article 110 warning systems

Hard parts and mitigations

OS background limits: permissions requested in peace; app must earn its place

Battery: adaptive duty cycle, target 8 h from 50%

Flooding: signed, rate-limited, portal aggregated

Privacy: pseudonyms, hop distance not GPS, L3 gossip off

Consent: published regulation, customer opt-out, relay isolated from home LAN

Residual risk: iOS background BLE limits, ISP and regulator uptake, trust in a single issuer

Design target (NFR-RES-07), not yet measured: with 20% router participation, an official alert reaches 95% of connected nodes within 5 minutes (cells down, power up).

Dual use is not new; the combination is

	Consumer mesh apps	LoRa meshes	Hotspot opening	Street-light paper	Peerby	Pomóc
Works without backhaul	Yes	Yes	No	Partly	No	Yes
Works without grid power	Yes	Battery	No	No	No	Yes (phones)
Uses deployed hardware	Phones	No	Yes	No	n/a	Phones + routers
Verified identity	No	No	No	No	Platform	State, offline
Reason to install pre-crisis	No	No	n/a	Yes	Yes	Yes
Alerts verifiable offline	No	No	No	No	No	Yes

Proposed path to pilot

- 1

Simulation
protocol, both modes, coverage view
- 2

District pilot
one ISP, volunteers, a drill declaration
- 3

Regulation + wallet
mObywatel module, open specification

Dual use is not new: Meuser et al. 2019 (street lights) and Peerby show it. To our knowledge, nothing combines state identity, a router + phone backbone and offline-verifiable alerts. Hotspot opening: 00000JAPAN, Xfinity.

Scope and disclosure. Done: concept, requirements and diagrams (prepared 3 Oct). Not built: code, simulation or demo; the protocol simulation is planned. Concept, research, documentation and this deck were developed with AI assistance (Claude Code); the team answers for all of it. Sources: docs/prior-art.md in the repository; Meuser et al. 2019 (arXiv 1908.10233); Comcast press releases 2022 and 2023; 00000JAPAN (SoftBank, Access Network); Peerby (TechCrunch 2013); Bridgefy; Alert RCB / mObywatel announcements 2025.